DipNDeep

FINTECH RISK INTELLIGENCE
& MACHINE LEARNING

The $100M Retail Lending Dilemma: Scaling Loan Books Without Bleeding Capital.

End-to-End Delinquency Prediction, Asymmetric Risk Optimization,
& Explainable Underwriting Simulator on 150,000 Retail Borrowers.

150,000 Profiles Audited **ROC-AUC: 0.8686 | KS-Stat: 57.84%** **8.2× Default Detection Lift**

Production Live XAI App

By Ganendra Pradipa

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Why a 93.3% Accuracy Model Can Bankrupt a Retail Bank.

The Paradox

In a dataset of 150,000 borrowers, the natural default rate is **only 6.68%.** A naive model that predicts "All Borrowers Will Repay on Time" achieves **93.32% accuracy.** However, this model identifies **0% of defaulting borrowers,** effectively guaranteeing a **total write-off** of the capital at risk!

The Asymmetric Loss Reality

Profit vs. Loss in the Lending Industry Is Never Symmetrical:

- Performing Borrower (+FP) : Generates a +10% interest margin ($1,000 profit)
- Defaulting Borrower (-FN) : Wipes out 100% of the loan principal ($10,000 total loss)

1 Defaulting Borrower Wipes Out the Profits of 10 Performing Borrowers at Once!

We replace accuracy with risk metrics tailored to the banking industry:

ROC-AUC **PR-AUC** **KS-Statistic** **Cost-Calibrated Threshold P = 0.56**

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Behind 150,000 Balance Sheets: 4 Empirical Behavioral Risk Signals.

1

THE 70% UTILIZATION CLIFF

Credit Utilization Tipping Point

- Credit utilization < 30% : Default rate 1.94% (Safe Zone)
- Credit utilization 70–100%: Default rate 17.72% (9× Higher)
- Over-limit utilization (> 100%): Default rate 37.25% (20× Higher)

Takeaway: Balances above 70% are an indicator of severe liquidity stress.

2

THE 30-DAY CASCADE

Early Warning Signals for Payment Delinquency

- 0 missed payments : Default rate 4.6%
- 1 payment 30–59 days late : Default rate 23.5% (A 5.1× Surge!)

Takeaway: Even minor delinquency is not a small anomaly, but an early gateway to financial distress.

3

THE AGE & ASSET MOAT

Age & Career Stability Gap

- Young borrowers (< 30 years) : Default rate 11.73%
- Older borrowers (60+ years) : Default rate 3.10% (3.8× Lower)

Takeaway: Asset accumulation and career stability provide a natural buffer against risk.

4

MORTGAGE COLLATERAL BUFFER

Real Estate Collateral Protection

- Borrowers with Home Mortgage/Real Estate Loans: Default rate 5.51%
- Borrowers without Home Mortgage Loans (Consumer Credit Only): Default rate 8.04% (31% Higher Risk)

Takeaway: Homeownership serves as an anchor for financial discipline.

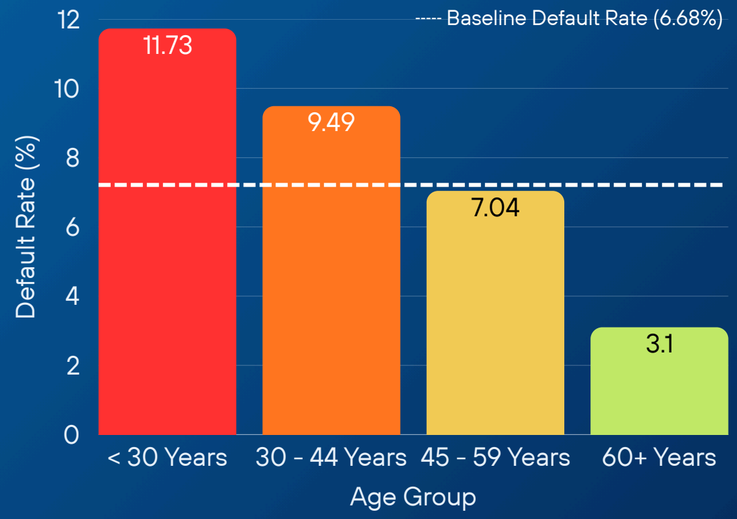
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Credit Risk Analytic : 4 Empirical Signal of Borrower Default

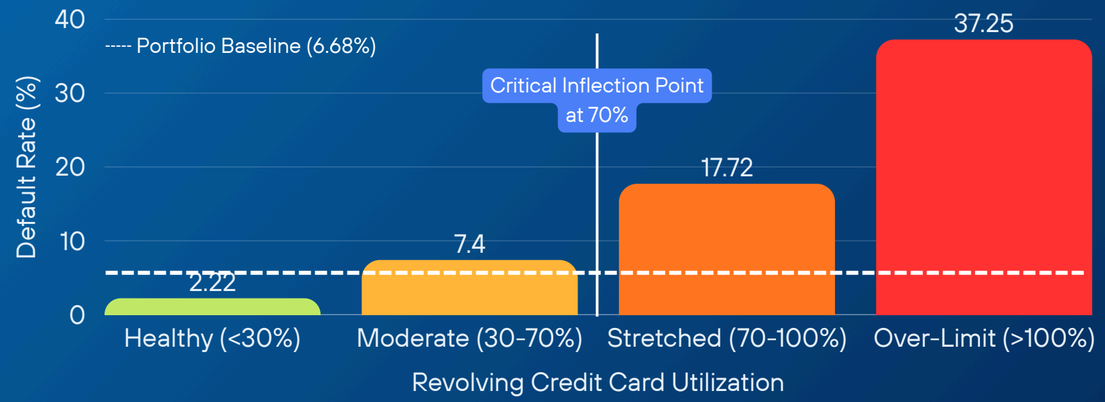
• The Age & Asset Gap

Default rates decline steadily with age, as income stabilty and asset accumulation increase over time.

3.8x
Risk Multiplier
< 30 Yrs vs 60+ Yrs



• The 70% Credit Utilization Tipping Point



• The Mortgage Collateral Buffer

Real estate and mortgage holding significantly reduce default risk by providing valuable collateral cushion.

No Real Estate Loan

8.3%
Default Rate

VS

Hold Real Estate / Mortgage

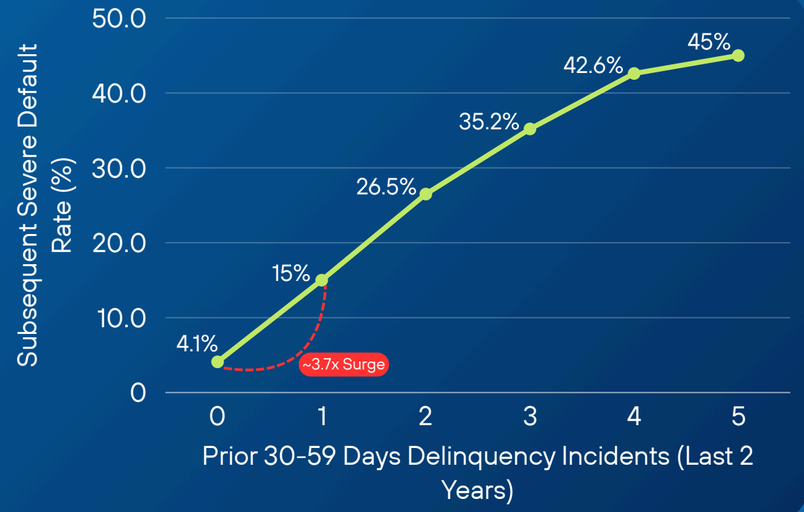
5.7%
Default Rate

-31% Lower Risk via Collateral Backing

----- Portfolio Baseline (6.68%)

• 30-Day Early Warning Cascade

A single 30–59 day delinquency isn't harmless: one past-due mark boosts two-year severe default risk from 4.1% to 15%, and up to 26.5% with two.



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Challenging the Industry Baseline: Classic Scorecard vs Tuned XGBoost.

TABEL BENCHMARK MODEL

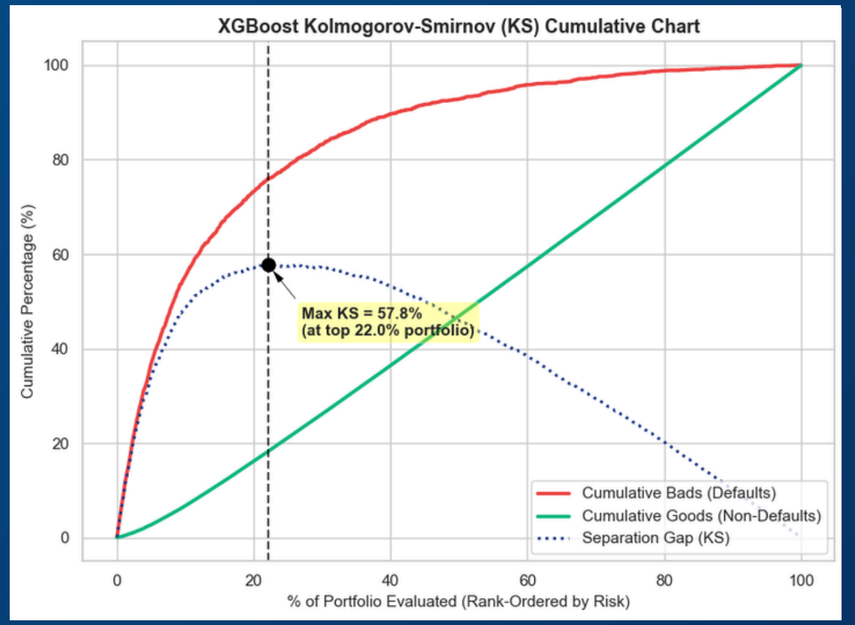
Model Name	ROC-AUC	PR-AUC	KS-Statistic	Top Decile Left
Logistic Regression	0.8034	0.2811	46.12%	5.8x Capture
Random Forest	0.8580	0.3802	55.40%	7.6x Capture
XGBoost	0.8686	0.4041	57.84%	8.2x Capture

ENGINEERING INNOVATION

- Imbalance Handling: scale_pos_weight ≈ 14.0
- 5 Engineered Anomaly Flags: Capture credit bureau data errors without data leakage
- KS Statistic 57.84%: Excellent discriminatory power (a KS score above 40% is considered excellent by banking industry standards)

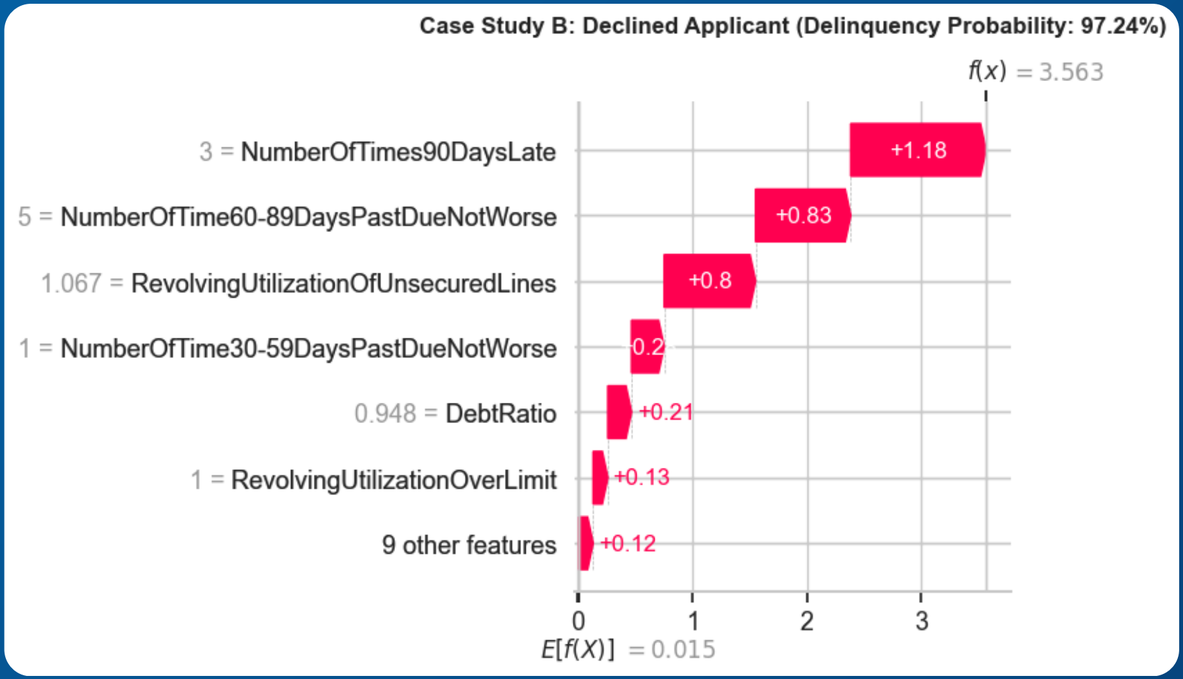
THE TOP DECILE BREAKTHROUGH

54.7% Delinquency Capture Rate: By screening only the top 10% of highest-risk applicants, our XGBoost model successfully identifies 54.7% of all defaulting borrowers in the market. (8.2× more effective than random screening).



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Cracking the Black-Box: Regulatory Compliance via Local SHAP XAI.

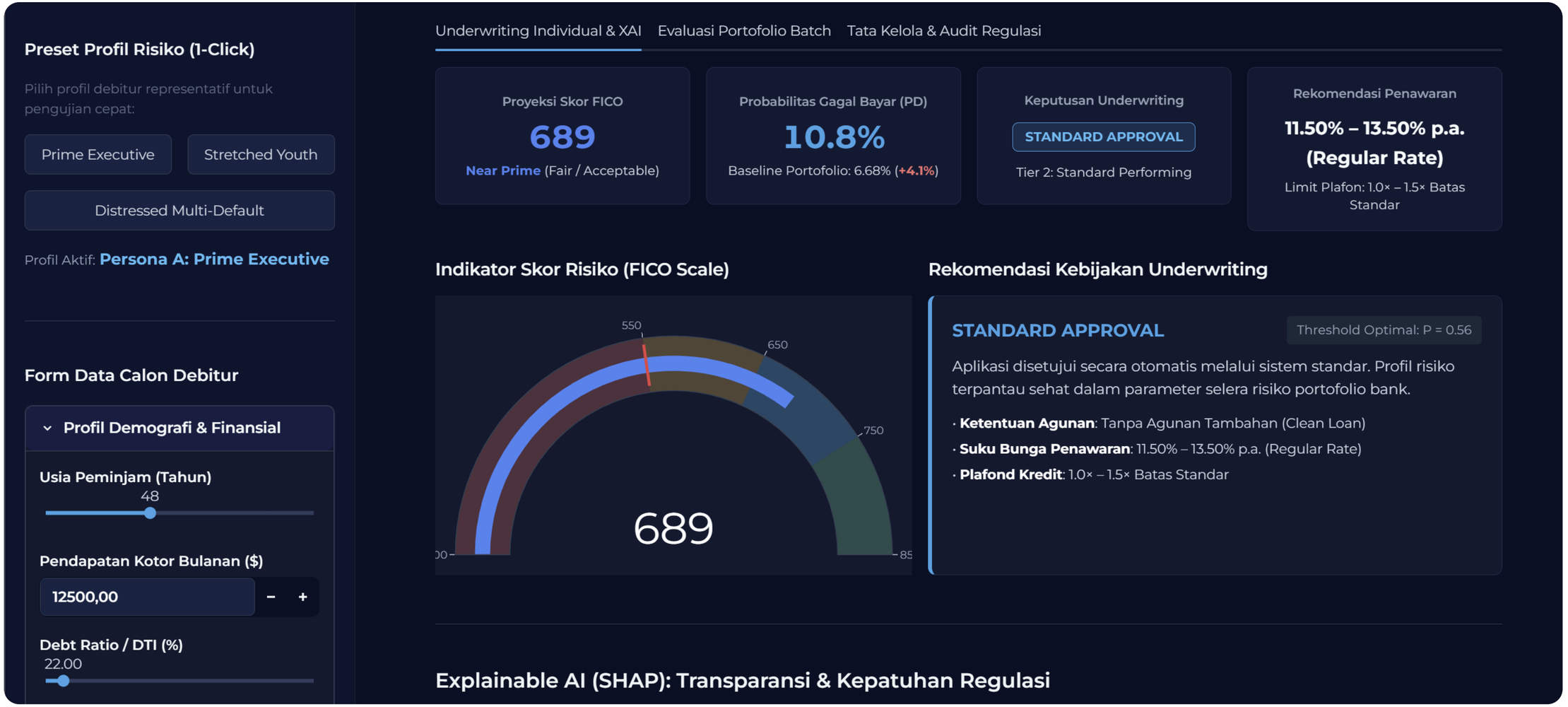


Formal Adverse Action Notice Reasons (Regulatory Audit Trail):

1. NumberOfTimes90DaysLate (Current Value: 3.00) -> SHAP Contribution: +1.180
2. NumberOfTime60-89DaysPastDueNotWorse (Current Value: 5.00) -> SHAP Contribution: +0.830
3. RevolvingUtilizationOfUnsecuredLines (Current Value: 1.07) -> SHAP Contribution: +0.797

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From Notebook to Production Cockpit: The Interactive Underwriting Simulator.



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Translating Algorithms to Dollar ROI: Preserving $3.55M Capital on a $100M Book.

Metrik Keuangan (Ours)	Traditional Heuristics	Logistic Scorecard	Tuned XGBoost
Default Detection (Recall)	35.0% (234 terdeteksi)	68.0% (454 deteksi)	82.5% (551 terdeteksi)
Missed Defaults (Loss FN)	434 nasabah macet	214 nasabah macet	Hanya 117 macet!
Bad Debt Loss ($10k / debt)	$4,340,000	$2,140,000	$1,170,000
Opportunity Cost ($1k / FP)	$1,200,000	$950,000	$820,000
Total Risk Cost	$5,540,000	$3,090,000	$1,990,000

$3,550,000 NET CAPITAL SAVED vs Traditional Heuristics

88.3% Healthy Portfolio Approval Rate Tetap Terjaga!

$1,100,000 NET CAPITAL SAVED vs Scorecard Klasik Basel II

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Summary

1. Predictive Superiority: ROC-AUC 0.8686 & KS-Statistic 57.84%
2. Regulatory Compliance: Full SHAP explainability & automated Adverse Action Notices
3. Quantified Financial Value: $3.55M loss reduction per $100M loan book



Let’s Connect & Collaborate

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